Slope, Motion & Comparing Rates

WARM UPS

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

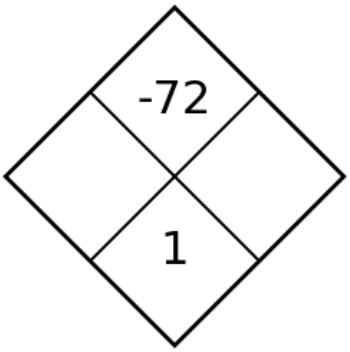
Solve for x. Show each step, then check your answer.

$$5(x-2)-2x=14$$

> **ANSWER / TEACHING NOTE**
> 5(x - 2) - 2x = 14 → 5x - 10 - 2x = 14 → 3x = 24 → x = 8.

2 | Diamond

Complete the diamond.



Slope, Motion & Comparing Rates

WARM UPS

3 | Spiral Review

A table includes (0,5), (2,11), (4,17). Find its slope.

ANSWER / TEACHING NOTE

m = (11 - 5)/(2 - 0) = 6/2 = 3.

4 | Spiral Review

Can {(1,8), (2,8), (2,9)} be a function? Explain.

ANSWER / TEACHING NOTE

No: input 2 produces two different outputs (8 and 9).

Slope, Motion & Comparing Rates

LEARNING FOCUS & NOTES

Learning focus | LG 02-ican-5 / LG 03-ican-3 / LG 03-ican-4

- I can calculate slope from two points or a table.
- I can interpret a slope using units and a direction.
- I can compare two rates fairly, including negative slopes.

Vocabulary | Highlight the important words

Rate of change

How much an output changes compared with an input change, with units attached.

Slope

The vertical change divided by the horizontal change of a straight line.

Positive / negative slope

A positive slope rises as x increases; a negative slope falls as x increases.

Zero slope

A horizontal line with no vertical change; slope zero.

Position

An object's location measured relative to a chosen reference point.

Velocity on a position-time graph

The rate and direction of change in position; the slope is signed and measured in distance per time.

Meaning & Method | Rewritten review notes

1. Always find slope with a ratio: (change in output) / (change in input), not the vertical change alone.

2. In context, read the sign and the units. A negative position-time slope can mean travel toward the chosen zero position; it does not mean negative speed.

3. To compare steepness or speed, compare magnitudes when the units are the same; explain the signs separately as direction or decreasing quantity.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Slope, Motion & Comparing Rates

EXAMPLES & YOU TRY ITS

Example

Find the cart's slope in meters per second. Explain how the sign describes its motion.

Time (s)	Position (m)
0	30
2	22
4	14

Calculate the slope in meters per second. Explain the sign and direction.

> **ANSWER / TEACHING NOTE**
> Δposition = 22 - 30 = -8 m; Δtime = 2 s; slope = -8/2 = -4 m/s. Position decreases by 4 m each second.

You Try It

Use this position-time table to calculate the cart's slope in m/s and explain its direction of motion.

Time (s)	Position (m)
0	10
3	25
6	40

Find slope with units and explain direction.

> **Find slope with units. What does its sign tell you?**
> Δposition = 25 - 10 = +15 m; Δtime = 3 s; slope = +5 m/s; position increases 5 m each second.

What does a positive or negative slope say about motion?

> **TEACHING NOTE**
> Positive slope means position increases over time; negative slope means position decreases relative to the reference direction.

Slope, Motion & Comparing Rates

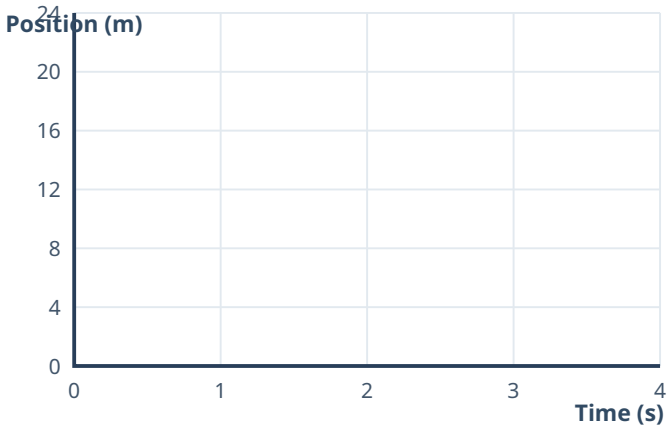
LAUNCH

Two walkers, one coordinate system

The values are positions measured in meters from the same starting marker. Graph both walkers using the given data, and explain how slope indicates direction.

Time (s)	Walker A (m)	Walker B (m)
0	0	20
1	4	17
2	8	14
3	12	11

Graph key: Walker A is blue; Walker B is red.



1 | Calculate each signed slope. Which walker changes position faster in magnitude? How do you know?

> **ANSWER / TEACHING NOTE**
> Walker A: +4 m/s. Walker B: -3 m/s. Walker A has greater speed magnitude, 4 m/s.

2 | What does the negative slope for Walker B mean? Why would it be inaccurate to call Walker B's speed -3 m/s?

> **ANSWER / TEACHING NOTE**
> B moves toward smaller coordinate values; its velocity is -3 m/s but speed is 3 m/s.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Slope, Motion & Comparing Rates

REVIEW & PREVIEW

1 ★ Position goes from 32 m at 0 s to 14 m at 6 s. Calculate average rate of change with units.

ANSWER / TEACHING NOTE

(14 - 32)/(6 - 0) = -18/6 = -3 m/s.

2 Find the slope between (-2,7) and (4,-5). Show Δy and Δx.

ANSWER / TEACHING NOTE

(-5 - 7)/(4 - (-2)) = -12/6 = -2.

3 ★ Two cyclists travel 30 m in 5 s and 42 m in 7 s. Compare their average rates.

ANSWER / TEACHING NOTE

Both are 6 m/s: 30/5 = 6 and 42/7 = 6.

Slope, Motion & Comparing Rates

REVIEW & PREVIEW

4 A tank starts at 70 gallons and loses 4 gallons per minute. Interpret the starting value and rate.

ANSWER / TEACHING NOTE

Initial volume 70 gallons; change -4 gallons each minute.

5 Spiral: For {(0,5),(2,5),(2,7)}, is it a function? List domain/range.

ANSWER / TEACHING NOTE

Not a function: x=2 has two outputs. Domain {0,2}; range {5,7}.

6 Preview: A scooter is 12 meters from a marker at time 0 and its position increases by 3 meters each second. Write an equation for position p after t seconds.

ANSWER / TEACHING NOTE

p = 3t + 12 (meters, seconds).

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.